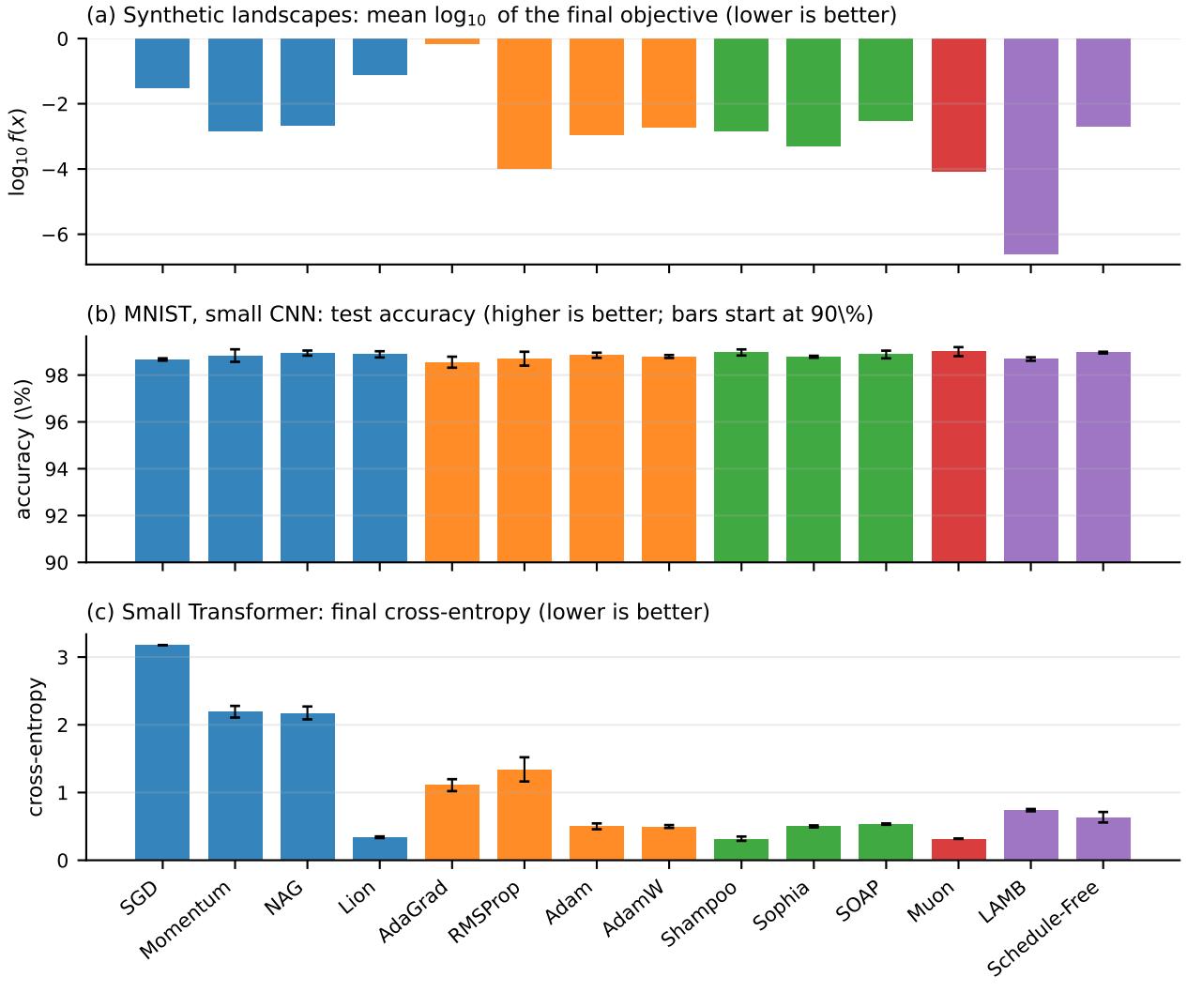
(a) Synthetic landscapes: mean log10 of the final objective (lower is better)
log10 f(x)
0
−2
−4
−6
(b) MNIST, small CNN: test accuracy (higher is better; bars start at 90\%)
accuracy (\%)
98
96
94
92
90
(c) Small Transformer: final cross-entropy (lower is better)
cross-entropy
3
2
1
0
SGD
Momentum
NAG
Lion
AdaGrad
RMSProp
Adam
AdamW
Shampoo
Sophia
SOAP
Muon
LAMB
Schedule-Free
F1 temporal filtering
F2 diagonal scaling
F3 matrix preconditioning
F4 spectral
F5 meta-structure